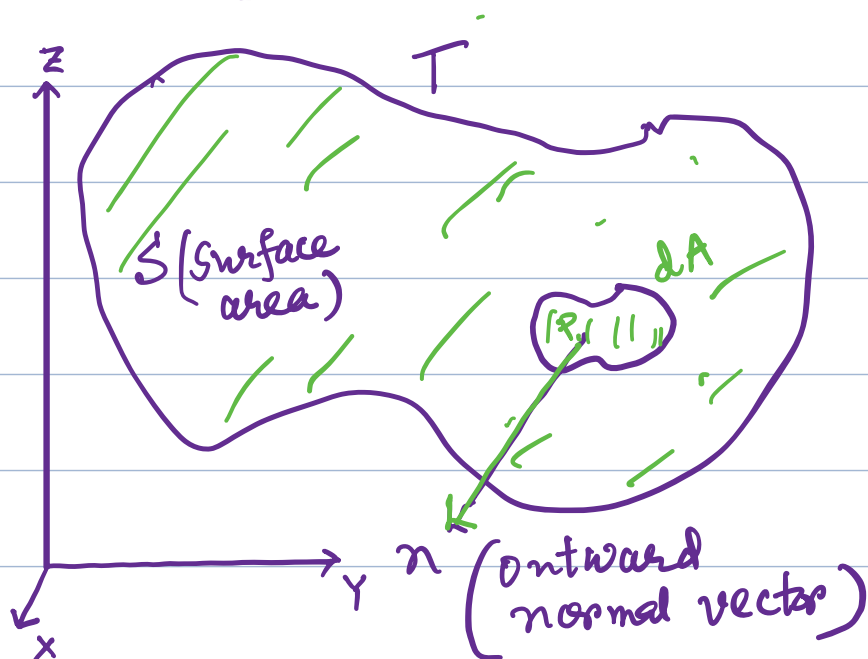


Addendum to Lecture 25

Heat Equation:

- Specific heat σ , density ρ of the material of the body are constant.
- Heat flows in the direction of decreasing temperature.



Let $u(x, y, z, t)$ is the temperature at a point (x, y, z) and time t .

Note that

$$\text{grad } u = i \frac{\partial u}{\partial x} + j \frac{\partial u}{\partial y} + k \frac{\partial u}{\partial z}$$

- Then velocity of heat flow in the body :

$$v = -k \text{ grad } u \quad \text{where } k = \text{Constant} \\ = \text{Thermal conductivity of the material}$$

- $v \cdot n$ is the component of v in the direction of n .
- The amount of heat leaving T or entering T per unit time at some point P of S through q

a small portion ΔS of area ΔA is $|v \cdot n \Delta A|$

Hence total amount of heat that flows across S

$$\text{from } T = \iint_S v \cdot n \, dA = -k \iiint_S (\text{grad } u) \cdot n \, dA$$

$$= -k \iiint_T \text{div}(\text{grad } u) \, dx \, dy \, dz \quad \left(\begin{array}{l} \text{By Gauss's} \\ \text{divergence} \\ \text{Theorem} \end{array} \right)$$

$$= -k \iiint_T \nabla^2 u \, dx \, dy \, dz \quad \dots \dots \dots \textcircled{1}$$

$$\left(\begin{array}{l} \text{Here } \text{div}(\text{grad } u) = \text{div} \left(i \frac{\partial u}{\partial x} + j \frac{\partial u}{\partial y} + k \frac{\partial u}{\partial z} \right) \\ \quad \quad \quad = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \nabla^2 u \end{array} \right)$$

On the other hand, the total amount of heat

$$\text{in } T : \quad H = \iiint_T \sigma \rho u(x, y, z, t) \, dx \, dy \, dz$$

$$\text{Now } - \frac{\partial H}{\partial t} = \iiint_T \sigma \rho \frac{\partial u}{\partial t} \, dx \, dy \, dz \quad \dots \dots \textcircled{2}$$

(time rate of decrease of H)

From $\textcircled{1}$ and $\textcircled{2}$

$$k \nabla^2 u = \sigma \rho \frac{\partial u}{\partial t}$$

$$\Rightarrow \frac{\partial u}{\partial t} = \frac{k}{\sigma \rho} \nabla^2 u$$

$$\Rightarrow \boxed{\frac{\partial u}{\partial t} = c^2 \nabla^2 u} \quad \text{where } c^2 = \frac{k}{\sigma \rho}$$

This is the Heat Equation.

1D Heat Equation:

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

2D Heat Equation:

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$